

RED DOG MINE 3 SSW, AK, USA

WMO: 701220

Lat: 68.0277N

Lon: 162.9212W

Elev: 942

StdP: 14.20

Time Zone: -9.00 (NAK)

Period: 10-19

WBAN: 26655

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.6	-21.2	-32.6	0.9	-25.8	-27.8	1.2	-20.3	25.3	8.7	23.1	11.5	2.9	N/A	0.776

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	73.0	56.5	68.9	54.8	65.5	53.5	58.3	68.6	56.7	65.6	55.1	63.1	5.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.4	65.3	59.7	52.6	60.9	58.8	50.7	56.8	57.1	25.6	67.6	24.5	66.0	23.5	63.1	64.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.5	15.0	DB	-26.5	79.1	7.3	4.3	-31.8	82.1	-36.1	84.6	-40.2	87.1	-45.5	90.2	(4)
(5)			WB	-26.5	61.8	6.7	1.5	-31.3	62.8	-35.3	63.7	-39.0	64.5	-43.9	65.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	6.6	8.2	6.9	20.2	36.6	50.9	53.1	48.3	39.3	26.8	12.9	5.7	(6)
(7)		DBStd	20.87	16.92	14.30	11.50	10.47	11.05	8.02	6.44	6.29	8.19	8.27	10.69	12.68	(7)
(8)		HDD50	8939	1345	1169	1337	893	430	87	38	106	328	719	1113	1374	(8)
(9)		HDD65	14099	1810	1589	1802	1343	881	427	372	518	771	1184	1563	1839	(9)
(10)		CDD50	322	0	0	0	0	14	113	135	53	7	0	0	0	(10)
(11)		CDD65	7	0	0	0	0	0	3	4	0	0	0	0	0	(11)
(12)		CDH74	87	0	0	0	0	3	43	41	0	0	0	0	0	(12)
(13)		CDH80	13	0	0	0	0	0	9	4	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.9	6.8	7.7	6.1	6.4	5.5	4.9	4.4	4.7	5.1	5.7	6.5	7.6	(14)
(15)	Precipitation	PrecAvg	17.8	1.3	1.4	1.0	0.8	0.7	0.8	1.9	2.8	2.0	2.4	1.5	1.2	(15)
(16)		PrecMax	24.8	2.7	4.4	5.0	2.2	1.8	1.5	3.9	6.1	4.4	4.6	4.6	3.1	(16)
(17)		PrecMin	12.8	0.3	0.2	0.2	0.1	0.0	0.4	0.5	0.6	0.5	0.9	0.2	0.4	(17)
(18)		PrecStd	3.1	0.6	1.0	0.9	0.6	0.5	0.3	0.8	1.3	0.8	0.9	1.0	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	35.5	33.7	46.0	69.8	79.7	79.8	71.3	61.6	52.2	37.8	35.4	(19)
MCWB			35.6	31.7	29.3	38.4	52.3	59.7	59.2	55.4	49.2	39.0	34.2	31.8	(20)	
2%		DB	33.2	31.7	30.9	40.1	62.6	73.5	73.2	65.9	57.6	45.5	34.1	32.6	(21)	
		MCWB	30.5	29.0	28.6	34.1	49.4	55.4	58.1	54.2	47.2	36.4	31.7	29.8	(22)	
5%		DB	30.6	29.2	27.5	36.8	57.0	69.1	69.2	61.8	54.2	41.0	31.2	28.9	(23)	
		MCWB	28.4	27.5	25.6	32.5	46.7	53.8	56.4	52.6	46.3	36.1	29.4	26.5	(24)	
10%		DB	27.7	26.8	24.2	34.0	51.8	65.7	65.5	58.6	50.9	37.8	27.9	24.3	(25)	
		MCWB	25.4	25.1	22.8	30.7	43.3	52.2	55.3	51.1	45.7	35.3	26.2	22.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.5	32.6	31.3	38.4	53.2	60.5	61.2	57.4	51.4	40.7	35.0	33.3	(27)
MCDWB			40.0	34.1	32.2	45.9	69.0	76.7	74.7	62.5	56.1	45.5	37.3	34.3	(28)	
2%		WB	31.4	29.5	28.5	34.5	49.6	57.2	59.4	56.0	49.9	38.6	32.1	30.1	(29)	
		MCDWB	32.8	30.9	30.0	39.1	62.4	69.5	70.4	62.8	54.1	42.6	34.0	32.6	(30)	
5%		WB	28.6	27.8	25.8	32.9	47.0	55.5	57.6	54.4	48.0	36.9	29.7	26.4	(31)	
		MCDWB	30.4	29.2	27.7	36.4	55.9	65.0	66.8	60.8	52.1	40.2	31.2	28.7	(32)	
10%		WB	25.5	25.3	22.8	31.1	44.3	53.6	56.1	52.3	46.4	35.2	26.2	23.0	(33)	
		MCDWB	27.5	26.7	24.6	34.2	51.0	63.2	64.3	57.2	50.4	38.4	27.7	24.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	11.5	11.5	11.6	13.8	18.2	14.9	12.6	11.7	8.4	8.6	10.1	(35)
MCDBR			8.7	9.9	11.2	12.1	21.7	23.9	22.4	17.5	15.2	10.0	7.3	9.5	(36)	
MCWBR			7.0	9.1	9.9	8.5	11.6	10.9	10.4	9.1	8.3	6.0	6.3	8.2	(37)	
5% WB		MCDBR	8.8	10.7	10.6	11.5	20.4	20.0	19.3	14.4	11.9	8.1	7.2	9.4	(38)	
		MCWBR	7.6	10.1	9.7	8.2	11.4	10.0	9.8	8.0	7.0	5.5	6.8	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.169	0.233	0.256	0.319	0.356	0.336	0.338	0.341	0.304	0.231	0.162	N/A	(40)	
taud		1.901	2.074	2.222	2.092	2.158	2.438	2.455	2.448	2.476	2.266	1.922	N/A	(41)		
Ebn at Noon		40	181	253	259	262	275	270	255	237	188	37	0	(42)		
Edh at Noon		3	15	23	36	39	30	29	26	19	12	2	0	(43)		
(44)	All-Sky Solar Radiation	RadAvg	13	172	721	1427	1738	1864	1498	1080	651	229	34	0	(44)	
(45)		RadStd	2	18	54	72	191	161	167	134	73	25	4	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page